IPEDS DATA QUALITY AUDIT

Academic Year 2024 — Five-Dataset Review

Framework:	DAMA DMBOK Six Data Quality Dimensions
Datasets Audited:	5 IPEDS Datasets (Enrollment, Finances, Graduation Rates, Directory, Characteristics)
Reporting Period:	Academic Year 2024 (IPEDS 2024 Survey Cycle)
Report Date:	April 6, 2026
Prepared For:	Senior Leadership / Data Governance Team
Classification:	Internal Use

Overall Data Quality Score Summary

8.8/10	**8.5/10**	**9.2/10**	**8.2/10**	**9.5/10**
DF1 Enrollment	DF2 Finances	DF3 Grad Rates	DF4 Directory	DF5 Characteristics

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1. Executive Summary

This report presents the findings of a comprehensive data quality audit conducted on five Integrated Postsecondary Education Data System (IPEDS) datasets for academic year 2024, published by the U.S. Department of Education's National Center for Education Statistics (NCES). The audit was structured using the DAMA Data Management Body of Knowledge (DMBOK) framework, evaluating six core dimensions of data quality: Completeness, Uniqueness, Timeliness, Validity, Accuracy, and Consistency.

Overall, the five datasets demonstrate a **high standard of data quality**, with average scores ranging from 8.2 to 9.5 out of 10. The datasets are well-structured, internally consistent, and aligned with IPEDS survey design conventions. Institutional identifiers (UNITID) are fully referentially intact across all five files — a significant governance strength.

The audit identified **21 specific data quality findings**, twelve requiring active governance remediation. The most significant issues are: (1) the Directory dataset (DF4) contains coordinate anomalies and a byte-order-mark encoding defect in the primary key; and (2) the Finances dataset (DF2) has a 13.3% imputation rate and 22.2% null rate on total revenues, reducing confidence in financial benchmarking.

Key Findings at a Glance

- DF5: Characteristics scored highest at **9.5/10** — exceptional completeness, valid coded fields, and full referential integrity.
- DF4: Directory scored lowest at **8.2/10** — coordinate outliers, a BOM encoding defect in the primary key, and elevated URL field missingness.
- DF1 and DF3 are multi-row by design (one row per institution-level combination). This is intentional architecture, not a data quality defect.
- DF2: Finances has 13.3% imputed values and 22.2% null total revenue. Financial benchmarking requires imputation disclosures.
- 14 institutions coded as 'closed' in the Directory still appear in the Enrollment dataset — a currency gap requiring reconciliation.
- All four transactional datasets achieve **100% referential integrity** against the Directory master register.

Recommendation for Senior Leadership

Priority Action: Establish a Formal IPEDS Data Governance Protocol

The five IPEDS datasets function as an integrated system. No single file contains a complete institutional profile. Leadership should direct the Data Governance Unit to: (1) designate DF4 as the authoritative institution master register; (2) resolve the BOM encoding defect in DF4 before the next analytical cycle; (3) establish a formal imputation disclosure policy for all financial reporting using DF2; and (4) exclude closed institutions (DEATHYR-flagged) from all active enrollment and performance analyses by default.

2. Audit Scope and Methodology

Datasets Audited

Dataset	Filename	Rows	Columns	Description
DF1	df1_enrollment_effy2024.csv	115,026	72	12-month enrollment by race and gender
DF2	df2_finances_f2324_f1a.csv	1,911	288	Institutional financial data (FY 2023-24)
DF3	df3_gradrate_gr2024.csv	51,011	66	Graduation rates by cohort, race, gender
DF4	df4_directory_hd2024.csv	6,072	72	Institution directory with location, control, type
DF5	df5_characteristics_ic2024.csv	5,963	111	Institution characteristics (programs, admissions)

DAMA DMBOK Framework

The audit applied the six data quality dimensions from the DAMA DMBOK (2nd Edition). Each dimension was evaluated through automated profiling, statistical analysis, cross-dataset joins, and domain-specific validation. Scores use a 0-10 scale where 10 represents full conformance with quality standards.

Dimension	Definition	Tests Applied
Completeness	Degree to which required values are present	Null rate analysis, mandatory field coverage
Uniqueness	Absence of unintended duplicate records	UNITID deduplication, full-row duplicate scan
Timeliness	Data reflects correct reporting period	Year field validation, closed-institution check
Validity	Values conform to formats, domains, and rules	Range checks, coded field validation, flag review
Accuracy	Data correctly represents real-world entities	Cross-field sums, coordinate validation, encoding
Consistency	Data is coherent across datasets and time	Cross-dataset UNITID alignment, convention review

3. Dataset Profiles

The following profiles summarize the structure and basic statistics of each dataset. All five share UNITID as the primary institution identifier and belong to the 2024 IPEDS survey cycle. DF4 (Directory) is the master institution register against which all other datasets are anchored.

DF1: 12-Month Enrollment (effy2024)

Rows	Columns	Unique Institutions	Completeness
115,026	72	5,861	94.9%

Multi-row structure: one record per institution-enrollment-level combination. Uses EFFYLEV codes (-2=N/A, 1=Undergrad, 2=Graduate, 4=Non-degree). 88.3% of rows carry EFFYLEV=-2, the IPEDS 'not applicable' sentinel. Four Guam enrollment columns are 88-98% null due to limited reporting.

DF2: Institutional Finances (f2324_f1a)

Rows	Columns	Unique Institutions	Completeness
1,911	288	1,911	93.4%

One record per institution. Public institutions under GASB accounting (Form F-1A). 288 columns include 142 X-prefix flag columns (R=Reported, A=Imputed, Z=Zero). 13.3% of flag cells are imputed. Six financial subcomponent columns exceed 58% null.

DF3: Graduation Rates (gr2024)

Rows	Columns	Unique Institutions	Completeness
51,011	66	3,684	96.9%

Multi-row structure: one record per institution-GRTYPE-CHRTSTAT combination. GRTOTLT and gender breakdowns are raw cohort counts, not percentages. Gender totals verified internally consistent (zero mismatches across all rows).

DF4: Institution Directory (hd2024)

Rows	Columns	Unique Institutions	Completeness
6,072	72	6,072	96.7%

One record per institution — the master reference file. Contains geographic, administrative, and classification fields. Key issues: BOM in UNITID header, 13 coordinate outliers, 2 encoding errors in institution names, 71 closed institutions (DEATHYR set).

DF5: Institution Characteristics (ic2024)

Rows	Columns	Unique Institutions	Completeness
5,963	111	5,963	99.4%

One record per institution. Binary and coded fields for programs, admissions, and athletics. Exceptionally clean: only DISABPCT has nulls (71.2%), all other categorical fields within valid domain ranges.

4. Dimension 1: Completeness

Completeness measures the degree to which all required data values are present. Evaluated at cell level (individual values) and column level (field-wide null rates).

8/10	7/10	9/10	8/10	9/10
DF1: Enrollment	DF2: Finances	DF3: Grad Rates	DF4: Directory	DF5: Characteristics

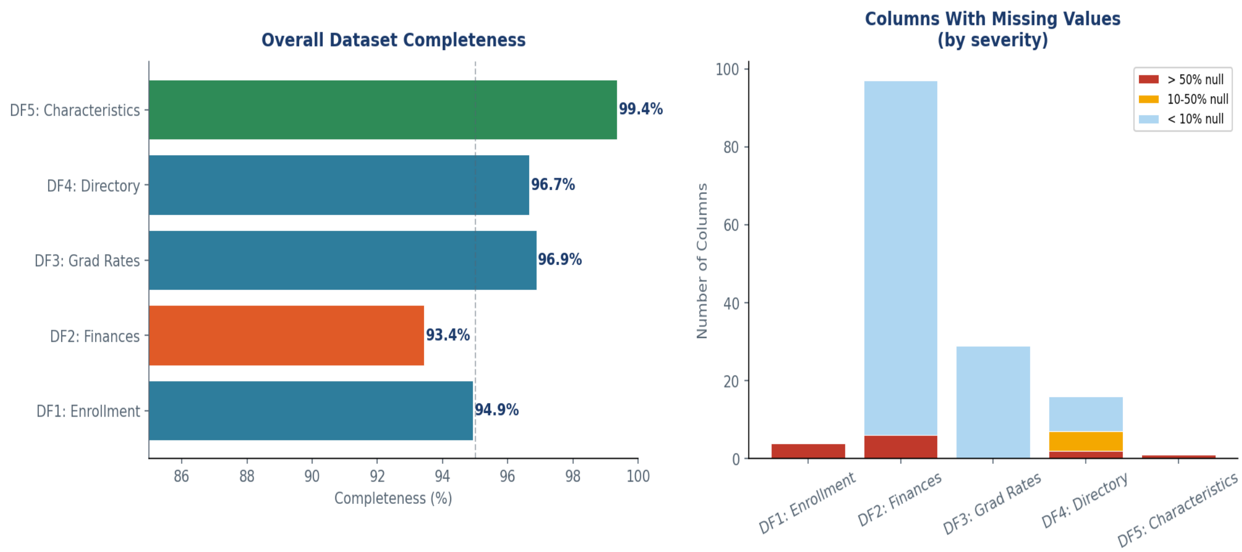

Figure 1: Overall completeness rates (left) and null severity distribution (right) across all five datasets.

Findings

F1.1 — DF2: Finances has 6 columns >58% null

Columns F1B06, F1B07, F1C121, F1C122, F1C131, F1C132 exceed 58% null. These sub-schedule fields apply only to a subset of public institutions. The missing data is structurally expected but should be documented in the data dictionary with population scope annotations.

F1.2 — DF1: Four Guam enrollment columns are 88-98% null

EFYGUUN, EFYGUAN, EFYGUTOT, EFYGUKN are near-fully null because very few institutions operate in Guam. Structurally expected, but these columns inflate null metrics and may mislead automated quality tools.

F1.3 — DF4: URL fields show significant missingness (14-72%)

Voluntary URL fields: ATHURL 71.8%, IALIAS 63.9%, VETURL 37.7% missing. The absence of athletic and veteran resource URLs is worth monitoring given public-facing accessibility implications.

F1.4 — DF5: DISABPCT has 71.2% null

> DISABPCT (% students with disabilities) is 71.2% null. Reported only by institutions meeting minimum enrollment thresholds. Not a data error, but limits disability equity analyses.

Recommendations

- **R1.1:** Annotate high-null columns with population scope in the data dictionary (e.g., 'Applicable only to Guam-territory institutions').
- **R1.2:** Establish completeness SLAs for mandatory vs. voluntary fields. Mark voluntary URL fields to avoid penalizing institutions in aggregate quality scoring.
- **R1.3:** Create a 'structural nulls' exclusion list for automated dashboards so expected-sparse fields do not distort composite scores.

5. Dimension 2: Uniqueness

Uniqueness measures the absence of unintended duplicate records. In multi-row datasets (DF1 and DF3), UNITID appearing multiple times per institution is intentional design. Uniqueness is evaluated relative to each dataset's intended grain.

10/10	10/10	10/10	9/10	10/10
DF1: Enrollment	DF2: Finances	DF3: Grad Rates	DF4: Directory	DF5: Characteristics

Findings

F2.1 — DF1, DF3: Multi-row structure is by design, not a defect

DF1 has 5,861 unique institutions across 115,026 rows (segmented by EFFYLEV enrollment level). DF3 has 3,684 unique institutions across 51,011 rows (segmented by GRTYPE/CHRTSTAT). No full-row duplicates were detected in any dataset. Governance documentation should explicitly describe the grain of each dataset to prevent misuse.

F2.2 — DF4: 167 institutions share duplicate names

167 records share institution names with at least one other institution (e.g., multiple 'ATA College' locations). These are legitimate multi-campus chains with distinct UNITID values. Name-based deduplication will produce incorrect results; UNITID must be the primary key for all joins.

Recommendations

- **R2.1:** Document dataset grain in the formal data dictionary: DF1 = UNITID x EFFYLEV; DF3 = UNITID x GRTYPE x CHRTSTAT; DF2/DF4/DF5 = UNITID.
- **R2.2:** Enforce UNITID-based joins at all downstream analytical pipelines. Prohibit name-based matching.

6. Dimension 3: Timeliness

Timeliness assesses whether data reflects the correct reporting period and is sufficiently current for its intended use. The 2024 cycle captures 2023-2024 academic year activity. Timeliness was evaluated by verifying alignment with expected reporting periods and identifying currency gaps.

8/10	8/10	8/10	9/10	9/10
DF1: Enrollment	DF2: Finances	DF3: Grad Rates	DF4: Directory	DF5: Characteristics

Findings

F3.1 — 14 closed institutions appear in DF1 enrollment data

DF4 records 71 institutions with DEATHYR (year closed) set. Of these, 14 still appear in DF1. Closed or merged institutions in enrollment data may inflate sector-level counts and confuse trend analyses.

F3.2 — No internal date-stamp fields in dataset records

None of the five datasets contain a field confirming when each record was last updated. Timeliness is inferred from the survey cycle suffix in filenames. This makes it difficult to distinguish genuinely current data from stale records.

F3.3 — DF4 contains one future-dated DEATHYR=2025

One institution in DF4 is recorded as closing in 2025, after the report date. This should be reviewed to confirm it represents an anticipated closure rather than a data entry error.

Recommendations

- **R3.1:** Exclude DEATHYR-flagged institutions from active analyses by default. Add a 'closed_institution' filter flag to all analytical views.
- **R3.2:** Add metadata audit timestamps to the pipeline. Record ingest date and last-update date for all IPEDS extracts.
- **R3.3:** Review the DEATHYR=2025 record with the source data provider.

7. Dimension 4: Validity

Validity assesses whether data values conform to defined formats, allowable domains, business rules, and structural conventions. Includes coded field dictionaries, sentinel values (-2 = 'not applicable'), flag patterns, and format standards.

9/10	8/10	9/10	7/10	10/10
DF1: Enrollment	DF2: Finances	DF3: Grad Rates	DF4: Directory	DF5: Characteristics

Findings

F4.1 — DF4: BOM character corrupts the UNITID column header [CRITICAL]

The UNITID column in DF4 carries a UTF-8 byte-order mark (BOM) prefix, rendering the header as 'ï»¿UNITID' when read with Latin-1 encoding. Any pipeline joining on UNITID from DF4 without handling this BOM will silently fail, producing empty join results. This is the highest-priority finding in this audit.

F4.2 — DF4: 13 coordinates fall outside all U.S. territory bounds

13 institution records have lat/lon values outside the bounding box covering all 50 states, D.C., Puerto Rico, Guam, and U.S. territories. Note: 172 records correctly reflect Alaska, Hawaii, and territories and are not defects.

F4.3 — DF4: 2 institution names contain garbled characters

Two Hispanic-serving institution names show corrupted non-ASCII characters due to UTF-8/Latin-1 encoding mismatch: 'Colegio de CinematografÃ\xada Artes y Television' and 'Dewey University-Juana DÃ\xadaz'. Correct names contain Spanish accented characters.

F4.4 — DF4: 22 records carry CONTROL=-3 and ICLEVEL=-3 (unclassified)

22 directory records carry -3 in classification fields. In IPEDS coding, -3 denotes 'not applicable'. These institutions should be reviewed to confirm whether classification data is genuinely unavailable.

F4.5 — DF2: Z-flag (zero suppression) in 8,069 cells may be misread

The Z flag indicates suppressed/zero values. Analytical users may treat suppressed zeros as true zero values, distorting revenue and expenditure aggregations.

Recommendations

- **R4.1:** Re-encode DF4 to UTF-8 and implement BOM-stripping as a mandatory preprocessing step for all IPEDS CSV imports.
- **R4.2:** Perform manual coordinate review for the 13 flagged institutions using NCES institution locator data.

• **R4.3:** Correct the 2 institution name encoding errors and add a character validation rule in the ingestion pipeline.

• **R4.4:** Document the Z-flag suppression rule. Mask Z-flagged cells as NULL in analytical views rather than allowing them to be read as 0.

8. Dimension 5: Accuracy

Accuracy measures how well data correctly represents real-world entities and events. Assessed through cross-field consistency checks, range validations, and checks for implausible or corrupted values.

10/10	8/10	9/10	7/10	9/10
DF1: Enrollment	DF2: Finances	DF3: Grad Rates	DF4: Directory	DF5: Characteristics

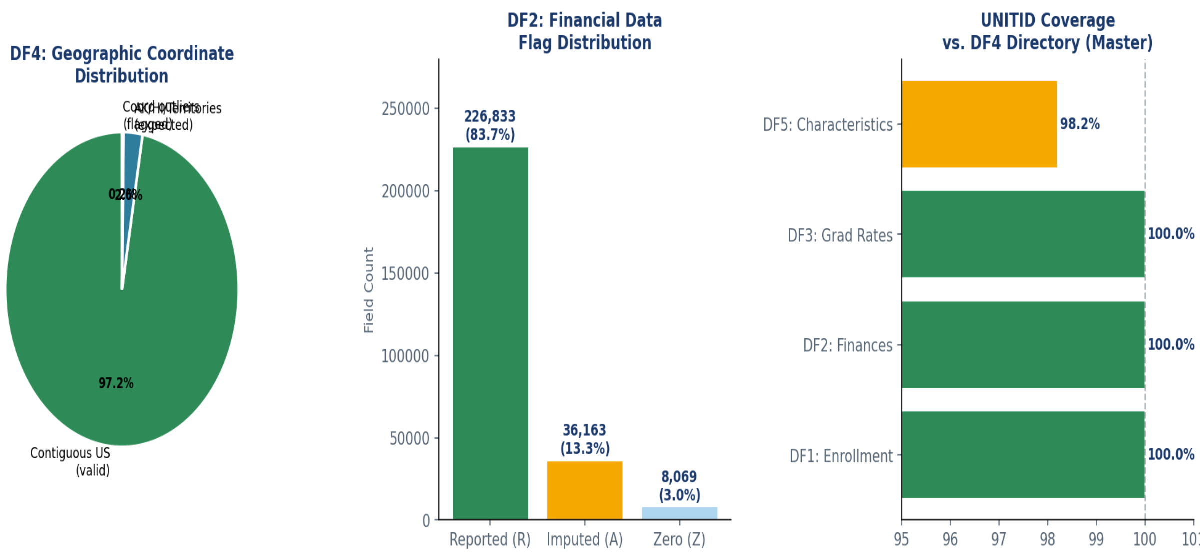

Figure 2: Coordinate distribution in DF4 (left), financial flags in DF2 (center), UNITID coverage vs. directory (right).

Findings

F5.1 — DF1 and DF3: Gender totals are internally accurate

Cross-field validation confirmed that reported totals in both DF1 and DF3 match exactly the sum of male and female sub-totals. Zero mismatches detected across all rows. This is a significant accuracy strength.

F5.2 — DF2: 22.2% of institutions have null total revenue

425 of 1,911 institutions (22.2%) have a null value for F1A01 (total revenues). The scale of this gap warrants investigation into whether these represent non-submitters, data processing gaps, or institutions not subject to Form F-1A.

F5.3 — DF2: 13.3% imputation rate across financial fields

13.3% of flag cells carry the 'A' (imputed) flag, meaning values were estimated rather than directly provided. Imputed values reduce reliability for institution-level financial benchmarking and must be disclosed in accountability reporting.

Recommendations

• **R5.1:** Investigate the 425 null F1A01 institutions. Separate non-submitters from institutions legitimately excluded from Form F-1A.

• **R5.2:** All DF2 financial analyses should include an imputation disclosure. Create 'high-confidence' analytical views excluding records with >20% imputed fields.

• **R5.3:** Add automated cross-field sum validation to the ingestion pipeline for DF1 and DF3 as a regression check.

9. Dimension 6: Consistency

Consistency measures whether data is coherent and compatible across related datasets and time periods. The primary test is referential integrity: every UNITID in a transactional dataset should resolve to a record in the Directory (DF4).

8/10	10/10	10/10	9/10	10/10
DF1: Enrollment	DF2: Finances	DF3: Grad Rates	DF4: Directory	DF5: Characteristics

Findings

F6.1 — 100% referential integrity across all four transactional datasets

Every UNITID in DF1, DF2, DF3, and DF5 resolves to a valid entry in DF4. Zero orphan records detected across 119,911 total records checked. This is an exceptional governance outcome and the strongest consistency finding in this audit.

F6.2 — DF5 covers 5,963 of 6,072 directory institutions (98.2%)

DF5 is missing 109 institutions present in DF4. These institutions did not submit characteristics survey data. The 1.8% gap is within normal non-response bounds but should be monitored for systematic patterns.

F6.3 — Sentinel value -2 used inconsistently across datasets

IPEDS code -2 ('not applicable') is used broadly in DF1 (88.3% of EFFYLEV values) but appears in DF5 PEO columns only for specific sub-populations. Without a centralized data dictionary defining -2 semantics per field, users may conflate 'not applicable' with missing data.

F6.4 — Encoding conventions are inconsistent across the five files

DF4 uses BOM-prefixed encoding that renders incorrectly under Latin-1, while the other four files read correctly under Latin-1. Inconsistent encoding increases ingestion complexity and risk of errors in analytical pipelines.

Recommendations

- **R6.1:** Record the 100% referential integrity result as a baseline in the governance register. Rerun at each IPEDS refresh cycle.
- **R6.2:** Investigate the 109 non-responding institutions in DF5 for systematic non-response bias.
- **R6.3:** Create a cross-dataset sentinel value registry documenting -2, -3, and Z codes per column per dataset.
- **R6.4:** Standardize all IPEDS extracts to UTF-8 without BOM at the ingestion layer. Add encoding validation to the pipeline.

10. Data Quality Scoring Summary

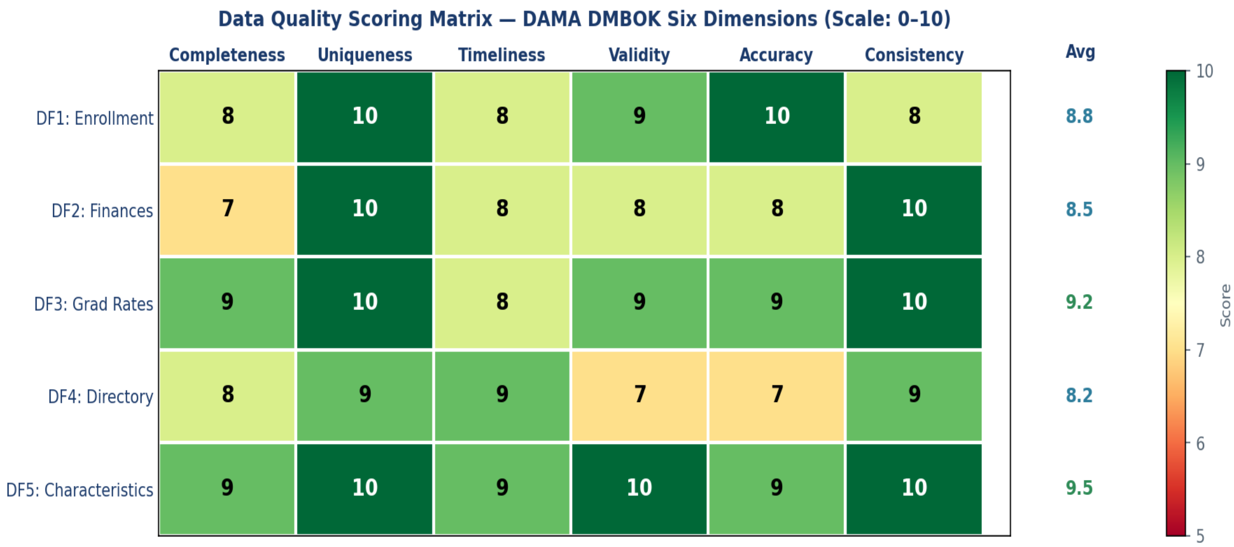

Figure 3: Data quality scoring matrix. Green = 9-10, Teal = 8, Yellow = 7.

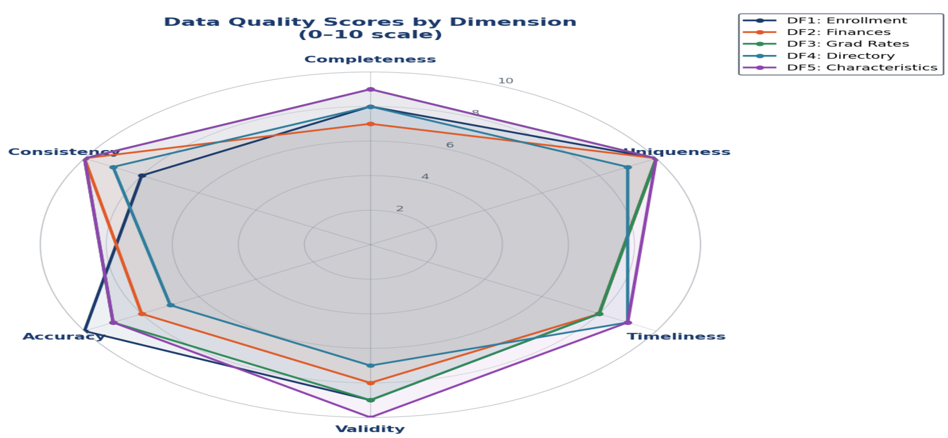

Figure 4: Radar chart comparing quality profiles across all five datasets.

Dataset	Completeness	Uniqueness	Timeliness	Validity	Accuracy	Consistency	Average
DF1: Enrollment	8	10	8	9	10	8	**8.8**
DF2: Finances	7	10	8	8	8	10	**8.5**
DF3: Grad Rates	9	10	8	9	9	10	**9.2**
DF4: Directory	8	9	9	7	7	9	**8.2**
DF5: Characteristics	9	10	9	10	9	10	**9.5**

Scores reflect automated profiling, rule-based validation, and cross-dataset analysis. A score of 10 represents full conformance with quality standards for the dataset's design intent. Scores are normalized for structural multi-row patterns in DF1 and DF3, which are by design and not defects.

11. Governance Recommendations

The following prioritized recommendations are drawn from all 21 findings across the six quality dimensions.

Priority	ID	Recommendation	Finding(s)	Dataset(s)
Critical	R4.1	Re-encode DF4 to UTF-8; strip BOM at ingest	F4.1	DF4
Critical	R3.1	Exclude DEATHYR-flagged closed institutions from active analyses	F3.1	DF4, DF1
Critical	R5.2	Add imputation disclosure to all DF2 financial analyses	F5.3	DF2
Critical	R5.1	Investigate 425 null revenue institutions in DF2	F5.2	DF2
High	R4.2	Review and correct 13 coordinate outliers in DF4	F4.2	DF4
High	R4.3	Correct 2 encoding errors in institution names	F4.3	DF4
High	R6.4	Standardize all IPEDS files to UTF-8 at ingestion	F6.4	All
Medium	R6.1	Baseline referential integrity check in governance register	F6.1	All
Medium	R2.1	Document grain of each dataset in formal data dictionary	F2.1	DF1, DF3
Medium	R6.3	Create cross-dataset sentinel value registry (-2, -3, Z)	F6.3	All
Medium	R1.1	Annotate high-null columns with population scope in metadata	F1.1-F1.4	DF1-DF5
Medium	R3.2	Add metadata audit timestamps to IPEDS data pipeline	F3.2	All

12. Appendix

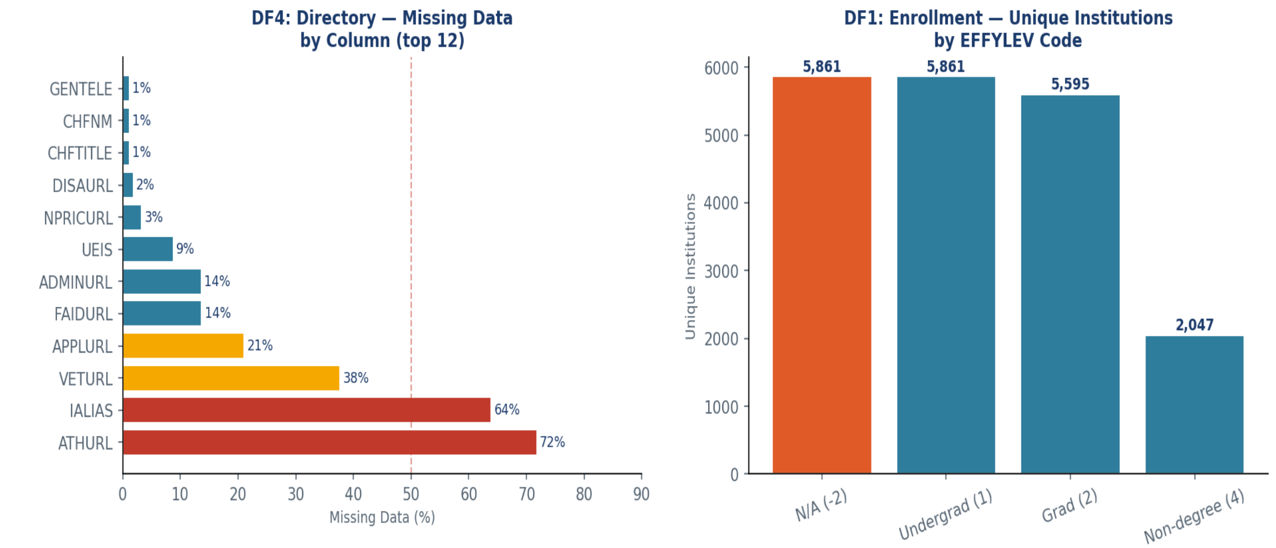

Figure 5: DF4 missing data by column (left). DF1 unique institutions by EFFYLEV code (right).

A. Complete Finding Reference Table

Finding	Dimension	Dataset	Description	Severity
F1.1	Completeness	DF2	6 financial sub-schedule columns >58% null	**Medium**
F1.2	Completeness	DF1	4 Guam enrollment columns 88-98% null	**Medium**
F1.3	Completeness	DF4	URL fields 14-72% null	**Medium**
F1.4	Completeness	DF5	DISABPCT 71.2% null	**Medium**
F2.1	Uniqueness	DF1, DF3	Multi-row grain is by design, not a defect	**Info**
F2.2	Uniqueness	DF4	167 institutions share duplicate names	**Medium**
F3.1	Timeliness	DF1, DF4	14 closed institutions in enrollment data	**High**
F3.2	Timeliness	All	No internal date-stamp fields in records	**Medium**
F3.3	Timeliness	DF4	Future DEATHYR=2025 on one record	**Medium**
F4.1	Validity	DF4	BOM character corrupts UNITID column header	**Critical**
F4.2	Validity	DF4	13 coordinates outside all U.S. territory bounds	**High**
F4.3	Validity	DF4	2 institution names with encoding errors	**Medium**
F4.4	Validity	DF4	22 records with CONTROL/ICLEVEL=-3	**Medium**
F4.5	Validity	DF2	8,069 Z-flag cells may be misread as zeros	**Medium**
F5.1	Accuracy	DF1, DF3	Gender totals verified internally consistent (PASS)	**Pass**
F5.2	Accuracy	DF2	22.2% of institutions have null total revenue	**High**

F5.3	Accuracy	DF2	13.3% imputation rate across financial fields	**High**
F6.1	Consistency	All	100% referential integrity — all datasets (PASS)	**Pass**
F6.2	Consistency	DF5	109 institutions missing from DF5 (1.8% gap)	**Medium**
F6.3	Consistency	All	Sentinel -2 semantics inconsistent across datasets	**Medium**
F6.4	Consistency	All	Encoding inconsistency: DF4 differs from others	**High**